HARDIK SRIVASTAVA

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Portfolio | LinkedIn | LeetCode | ResearchGate | Google Scholar | GitHub

EDUCATION

University of Washington — Seattle, WA
Master of Science in Data Science — Sept 2025 - Mar 2027

SRM Institute of Science and Technology — Chennai, TN
Bachelor of Technology in Computer Science with specialization in Big Data Analytics (GPA - 3.82/4) — May 2019 - Jun 2023

WORK EXPERIENCE

Graduate Research Assistant (P.I. - Prof. Zaid Harchaoui) — Sept 2025 - Present
Paul G. Allen School of Computer Science, University of Washington — *Seattle, WA*

- Engineered and fine-tuned **modular LoRA adapters** for **controllable text style transfer**, then built **metric-driven agents** to anonymize text via learned style representations, **constrained decoding** and attribution classifiers under privacy–fidelity constraints.
- Formalized **16+ writing styles** as interpretable control variables with **closed-loop logic** for dynamic perturbation scaling. Improved anonymity by **40%** on AuthorMix (30K texts/14 authors) while preserving **88%** semantic similarity and **5%** perplexity drift.
- Optimized agent policies via **multi-objective feedback** and conducted empirical analysis across multilingual datasets. Benchmarked the system against larger LLM baselines, achieving **privacy gains** at **3x lower inference cost** and **34% fewer generated tokens**.

Applied Scientist 2 — Jun 2023 - Aug 2025
JPMorgan Chase & Co. — *Hyderabad, TS*

- Led **fraud detection** initiatives saving over **$220M annually** by developing high-recall **XGBoost** and **CatBoost**; optimized recall–precision tradeoffs using **constrained objective tuning** while maintaining feature interpretability for stakeholders.
- Productionized Semi-Supervised Graph Neural Networks across 60M+ transaction nodes; introduced **topology-aware edge sampling** and **dynamic neighborhood aggregation**, boosting fraud-detection **precision** by **9.2%** over **2M transactions/day**.
- Architected a GPU-served **contrastive pre-trained Transformer** (ViT + BERT-style decoder), scaling to **500K+** requests/day, for OCR-based check fraud detection. Improved fraud precision by **19%** while reducing latency via batched mixed-precision inference.
- Built a **Learning-to-Rank** entity matching system over **3B+** KYC records using Pairwise Ranking loss to flag fraud entities; Outperformed the existing TF-IDF baselines, cutting **62% false positives** and saving **700+** analyst review hours daily in production.
- Developed **distributed data pipelines** in PySpark to handle terabyte-scale transactions. Performed **optimal feature selection** using **Decision Trees** that boosted **Fraud Capture Rate** by **14%** and enhanced predictive accuracy for risk scoring models.

Research Intern — Jun 2022 - Sept 2022
McGill University - *Mitacs Scholar* — *Montreal, QC*

- Developed **RoBERTa-based sentence embeddings** on AAC image captions to capture word-context. Achieved Krippendorff's alpha reliability score of **0.82** for context-aware vocab retrieval; deployed the model in ClickAAC mobile application.

Machine Learning Engineer - Intern — Jan 2022 - Jun 2022
Samsung Research — *Bangalore, KN*

- Optimized **word-context relationships** for Samsung Bixby search using bipartite graphs, reducing **false-positive** preds by **12%**. Matched **FP32** perf for INT8 inference using quantization-aware training for word-disambiguation task with a **2.9%** F1 gain.

PROJECTS

Agentic Ad Intelligence Engine [code] Built a hyperpersonalized ad intelligence system that constructs **dynamic user personas** from behavioral signals and generates **context-aware ad variants** conditioned on audience attributes; implemented **feedback-driven ranking** and **attention-aware decision logic** using ensemble engagement predictions for **smart ad delivery** enabling scalable personalization.

PUBLICATIONS

- Srivastava, H. (2023). Multi-Modal Sentiment Analysis Using Text and Audio for Customer Support Centers
- Srivastava, H. (2023). Neural Text Style Transfer with Custom Language Styles for Personalized Communication Systems
- Srivastava, H. (2021). Using NLP Techniques for Enhancing Augmentative and Alternative Communication Applications

ACCOMPLISHMENTS

- **Certifications** — AWS Certified Machine Learning Specialist, AWS Certified Cloud Practitioner
- **Hackathons** — xAI Hackathon'25 (**Bronze**), UWxDatabricks Hack'25 (**Bronze**), JPMC AWS DeepRacer'23 (**Gold**)

SKILLS

AI/Research	Machine Learning, Natural Language Processing, Generative AI, Sequence Modeling, Graph-ML, Transformers Ranking, Agentic-AI, Large Language Models, Time Series Forecasting, Data Analysis
CS/Tooling	C++, Python, Java, SQL, PyTorch, TensorFlow, Sklearn, Spark, Kafka, W&B, HuggingFace, Optuna, Git
Infra/MLOps	AWS, Docker, Kubernetes, Flask, MLFlow, RAG, LangChain, NVIDIA Triton, Feature Stores, Jenkins